

● MEDIUM

● INFORMATION AND IDEAS

Command of Evidence

03

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

The Post Office is a 1912 play by Rabindranath Tagore, originally written in Bengali. The character Amal is a young boy who imagines that the people he sees passing the window of his home are carefree even when engaged in work or chores, as is evident when he says to the daughter of a flower seller, _____blank

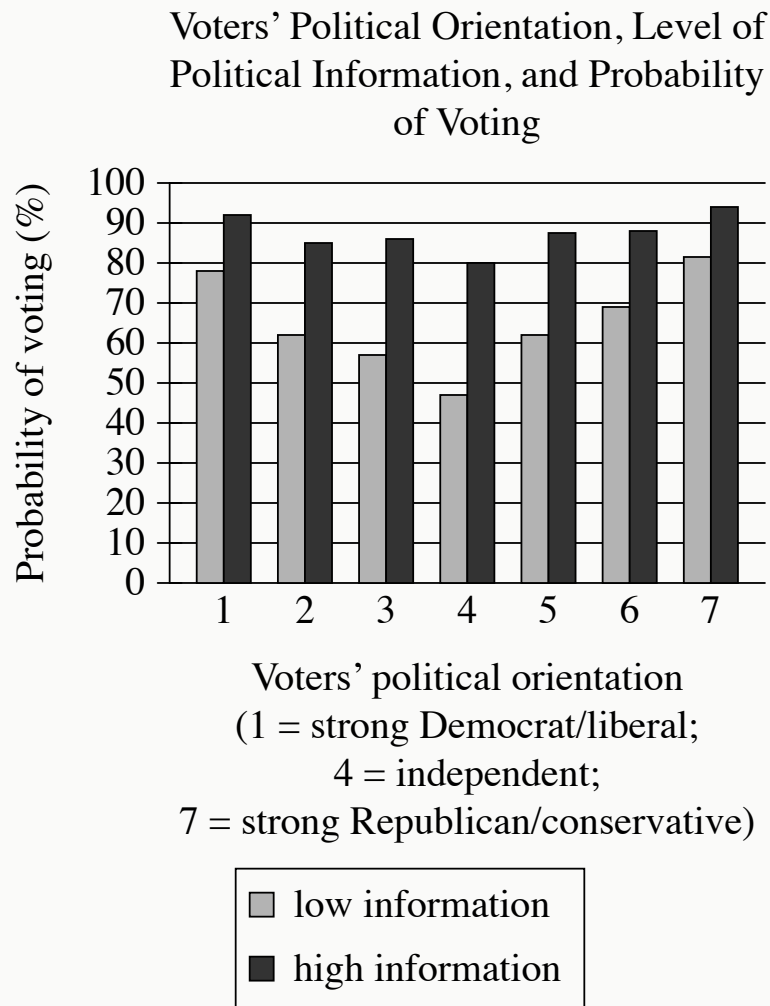
Which quotation from *The Post Office* most effectively illustrates the claim?

- A. “I see, you don’t wish to stop; I don’t care to stay on here either.”
- B. “Oh, flower gathering? That is why your feet seem so glad and your anklets jingle so merrily as you walk.”
- C. “I’ll pay when I grow up—before I leave to look for work out on the other side of that stream there.”
- D. “Wish I could be out too. Then I would pick some flowers for you from the very topmost branches right out of sight.”

“Looking Back on Girlhood” is an 1892 short story by Sarah Orne Jewett. In the story, the narrator explains that she prefers her hometown to other places she has visited: _____ blank

Which quotation from “Looking Back on Girlhood” most effectively illustrates this claim?

- A. “There is always something fresh, something to be traced or discovered, something particularly to be remembered.”
- B. “Two large rivers join just below the village at the head of tide-water, and these, with the great inflow from the sea, make a magnificent stream, bordered by lovely green fields that slope gently to long lines of willows at the water’s edge.”
- C. “I have had a good deal of journeying in my life, and taken great delight in it, but I have never taken greater delight than in my rides and drives and tramps and voyages within the borders of my native town.”
- D. “There is never-ending pleasure in making one’s self familiar with such a region.”



- For each data category, the following bars are shown:
 - low information
 - high information
- The data for the 7 categories are as follows:
 - 1:
 - low information: 78%
 - high information: 92%
 - 2:
 - low information: 62%
 - high information: 85%

- 3:
 - low information: 57%
 - high information: 86%
- 4:
 - low information: 47%
 - high information: 80%
- 5:
 - low information: 62%
 - high information: 87.5%
- 6:
 - low information: 69%
 - high information: 88%
- 7:
 - low information: 81.5%
 - high information: 94%

Economists Kerwin Kofi Charles and Melvin Stephens Jr. investigated a variety of factors that influence voter turnout in the United States. Using survey data that revealed whether respondents voted in national elections and how knowledgeable respondents are about politics, Charles and Stephens claim that the likelihood of voting is driven in part by potential voters' confidence in their assessments of candidates—essentially, the more informed voters are about politics, the more confident they are at evaluating whether candidates share their views, and thus the more likely they are to vote.

Which choice best describes data in the graph that support Charles and Stephens's claim?

- A. At each point on the political orientation scale, high-information voters were more likely than low-information voters to vote.
- B. Only low-information voters who identify as independents had a voting probability below 50%.
- C. The closer that low-information voters are to the ends of the political orientation scale, the more likely they were to vote.

- D. High-information voters were more likely to identify as strong Democrats or strong Republicans than low-information voters were.

Q4

INFORMATION AND IDEAS

Jan Gimsa, Robert Sleigh, and Ulrike Gimsa have hypothesized that the sail-like structure running down the back of the dinosaur *Spinosaurus aegyptiacus* improved the animal's success in underwater pursuits of prey species capable of making quick, evasive movements. To evaluate their hypothesis, a second team of researchers constructed two battery-powered mechanical models of *S. aegyptiacus*, one with a sail and one without, and subjected the models to a series of identical tests in a water-filled tank.

Which finding from the model tests, if true, would most strongly support Gimsa and colleagues' hypothesis?

- A. The model with a sail took significantly less time to complete a sharp turn while submerged than the model without a sail did.
- B. The model with a sail displaced significantly more water while submerged than the model without a sail did.
- C. The model with a sail had significantly less battery power remaining after completing the tests than the model without a sail did.
- D. The model with a sail took significantly longer to travel a specified distance while submerged than the model without a sail did.

The Land of Enchantment is a 1906 travel book by Lilian Whiting. In the book, which describes the experience of traveling through the southwestern United States by train, Whiting reflects on the escape from everyday life that such a journey provides: _____ blank

Which quotation from *The Land of Enchantment* most effectively illustrates the claim?

- A. “The opportunities and advantages already offered and constantly increasing are greater than would at first be considered possible.”
- B. “The social and the picturesque charm of the long journey is singularly enhanced by the leisurely stops made for refreshment.”
- C. “The real journey begins, of course, at Chicago, and as these trains leave in the evening the traveller fares forth in the seclusion of his berth.”
- D. “One experiences a certain sense of detachment from ordinary day and daylight duties that is exhilarating.”

Male túngara frogs make complex calls to attract mates, but their calls also attract frog-biting midges, insects that feed on the frogs' blood. Researchers Ximena Bernal and Priyanka de Silva wondered if the calls alone are sufficient for midges to locate the frogs or if midges use carbon dioxide emitted by frogs as an additional cue to their prey's whereabouts, like mosquitoes do. In an experiment, the researchers placed two midge traps in a túngara frog breeding area. One trap played recordings of túngara frog calls and the other released carbon dioxide along with playing the calls. Bernal and de Silva concluded that carbon dioxide does not serve as an additional cue to frog-biting midges.

Which finding from the experiment, if true, would most directly support Bernal and de Silva's conclusion?

- A. Only a small number of midges were found in the traps, though the majority were found in the trap that played calls and released carbon dioxide.
- B. Midges entered the trap that released carbon dioxide and played calls only during or immediately after periods of carbon dioxide release.
- C. More midges were found in the trap that only played calls than in the trap that played calls and released carbon dioxide.
- D. The trap that released carbon dioxide and played calls attracted few midges when carbon dioxide concentrations were low but attracted many midges when carbon dioxide concentrations were high.

Properties of Select Rotating Radio Transients

Name	Right ascension (hours)	Period (seconds)	Frequency (hertz)
J0545-03	5:45	1.074	0.931
J1654-2335	16:54:03	0.545	1.834
J0103+54	1:03:37	0.354	2.822
J0121+53	1:21	2.725	0.367
J0614-03	6:15	0.136	7.353

A student is researching rotating radio transients (RRATs), a subclass of pulsar stars characterized by short pulses of radio waves. The time between consecutive pulses of an RRAT is referred to as a period. Looking at the table, the student determines that _____blank

Which choice most effectively uses data from the table to complete the statement?

- A. J0614-03 has the shortest amount of time between consecutive pulses of all the RRATs in the table.
- B. J0545-03 and J0121+53 have the same amount of time between consecutive pulses.
- C. J1654-2335 has the longest amount of time between consecutive pulses of all the RRATs in the table.

- D. J0103+54 and J0121+53 both have more than one second of time between consecutive pulses.

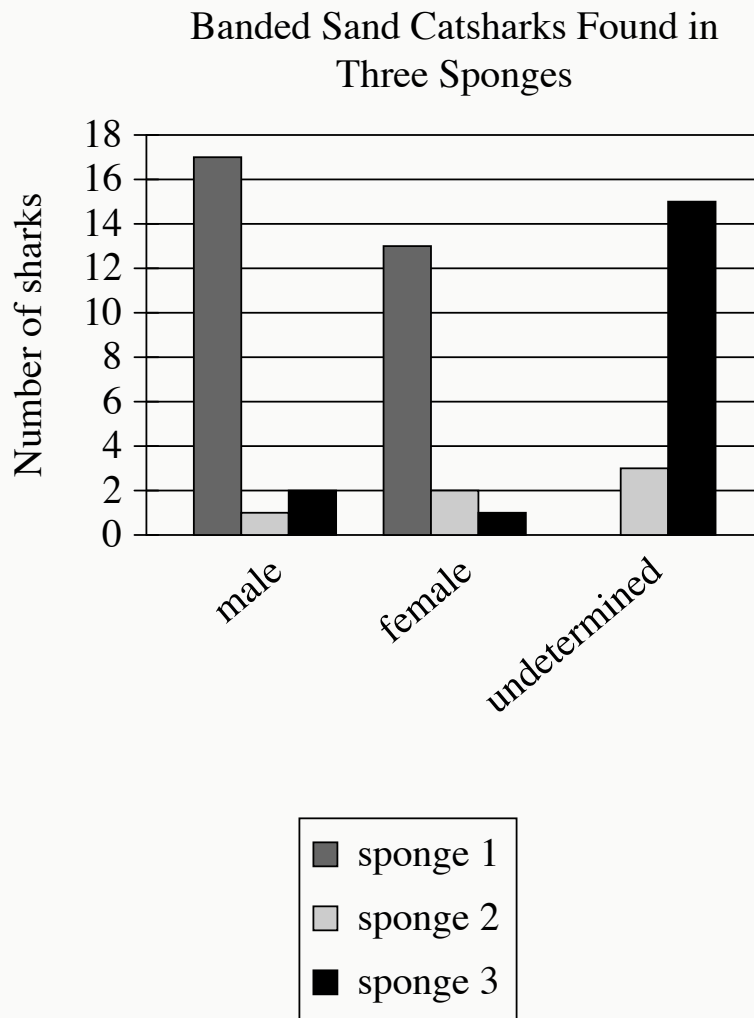
Q8

INFORMATION AND IDEAS

Plants like potatoes, tomatoes, and soybeans are susceptible to bacterial wilt disease caused by the bacteria *Ralstonia solanacearum*. A multinational team of scientists led by Zhong Wei studied whether other microbes in the soil might influence the degree to which plants are affected by the disease. The team sampled soil surrounding individual tomato plants over time and compared the results of plants that became diseased with those that remained healthy. They concluded that the presence of certain microbes in the soil might explain the difference between healthy and diseased plants.

Which finding, if true, would most directly support the team's conclusion?

- A. The soil surrounding healthy plants contained significantly higher concentrations of microbes known to inhibit *Ralstonia solanacearum* than the soil surrounding diseased plants did.
- B. The soil surrounding the plants contained high concentrations of *Ralstonia solanacearum* regardless of whether the plants were affected by wilt disease.
- C. The soil surrounding healthy plants tended to have significantly higher moisture levels than the soil surrounding diseased plants did.
- D. By the end of the experiment, over half the plants had been affected by wilt disease regardless of differences in the types and concentrations of microbes in the surrounding soil.



- For each data category, the following bars are shown:
 - sponge 1
 - sponge 2
 - sponge 3
- The Number of sharks data for the 3 categories are as follows:
 - male:
 - sponge 1: 17 sharks
 - sponge 2: 1 shark
 - sponge 3: 2 sharks
 - female:

- sponge 1: 13 sharks
- sponge 2: 2 sharks
- sponge 3: 1 shark
- undetermined:
 - sponge 1: 0 sharks
 - sponge 2: 3 sharks
 - sponge 3: 15 sharks

Researcher Helen O'Neill and her team were studying sponges in the family Irciniidae when they discovered that three of the sponges were functioning as microhabitats for banded sand catsharks (*Atelomycterus fasciatus*). Additionally, the team found that male and female *A. fasciatus* share the same sponge (the same microhabitat), a behavior not found among Port Jackson sharks (*Heterodontus portusjacksoni*), another seabed-dwelling species.

Which choice best describes data from the graph that support the underlined claim?

- A. Each of the three sponges was inhabited by at least one male *A. fasciatus* and at least one undetermined *A. fasciatus*.
- B. Sponge 1 was inhabited by 17 male *A. fasciatus*.
- C. Each of the three sponges was inhabited by at least one male *A. fasciatus* and at least one female *A. fasciatus*.
- D. Sponge 1 was inhabited by 13 female *A. fasciatus*.

In the 1970s, a roughly 60,000-year-old piece of hyena bone marked with nine notches was discovered at a site in western France once inhabited by Neanderthals. Although many believe that only modern humans developed systems for notating numbers, one archaeologist asserts that this artifact may be a sign that Neanderthals also recorded numerical information. The notches on the bone are unevenly spaced but approximately parallel, and microscopic analysis reveals that they were made with a single stone tool; according to the archaeologist, this suggests that the notches were all made at one time by one individual as a means of counting something.

Which finding, if true, would most directly weaken the underlined claim?

- A. Parallel lines are a common feature in modern humans' early systems for recording numerical information.
- B. More than nine approximately parallel notches made with a different stone tool are present on another artifact found at a site in western France.
- C. It would have taken careful effort to make evenly spaced lines on bone with the stone tools typically used by Neanderthals.
- D. Decorative art discovered at another Neanderthal site in western France primarily features patterns of unevenly spaced parallel lines.